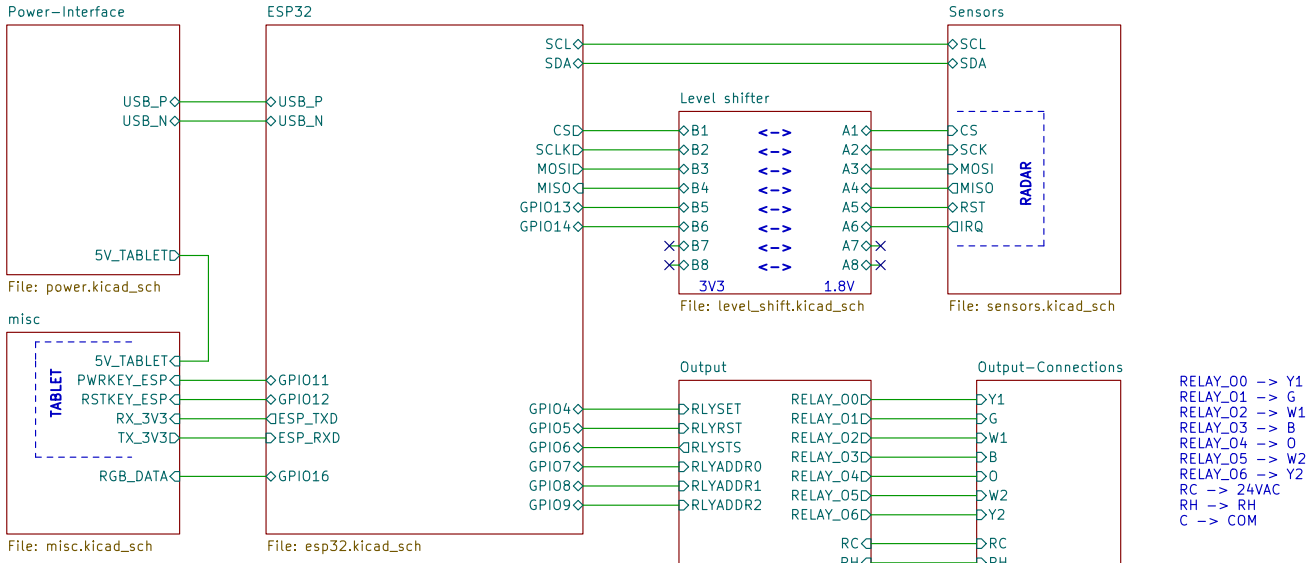


Relays, depending on the channel, switch between COM and 24VAC or RH wire

Y, G, O are switched betwen COM and 24VAC (RC)
W, B are switched between COM and RH wire

If C-wire is absent, 24VAC is absent, only W and B are switched to RH wire.

Mechanical

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial

N1 Housing

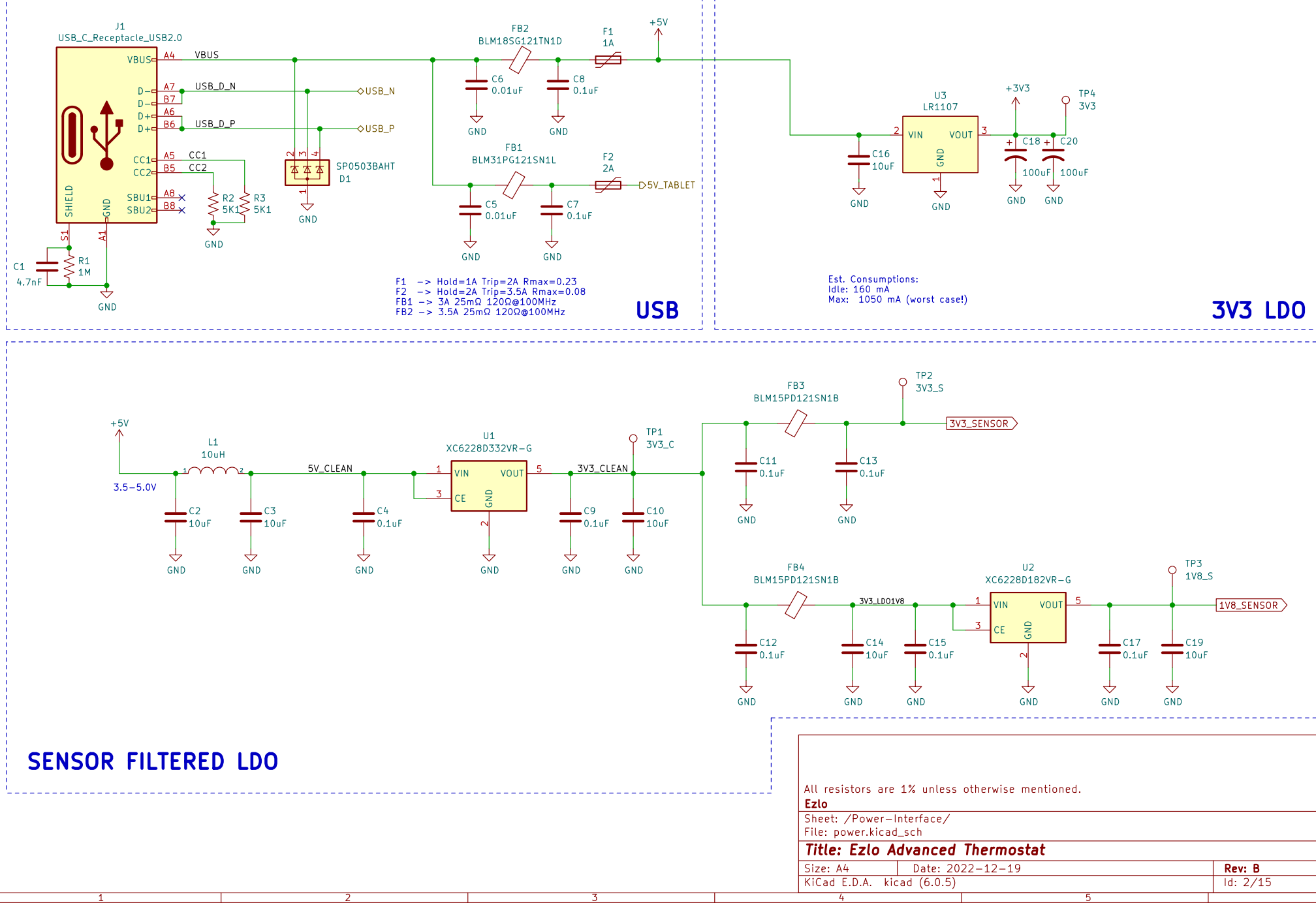
All resistors are 1% unless otherwise mentioned.

Ezlo

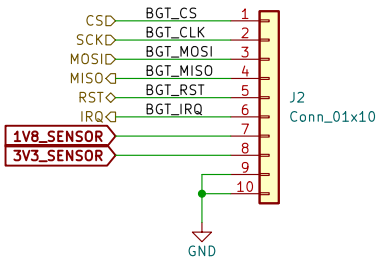
Sheet: /
File: adv thermostat.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 1/15

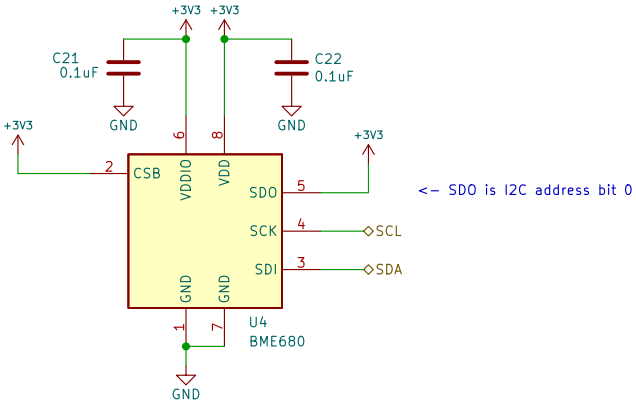

J1
USB_C_Receptacle_USB2.0
VBUS
USB_D_N
USB_D_P
CC1
CC2
SHIELD
GND
SBU1
SBU2
R1 1M
C1 4.7nF
R2 5K1
R3 5K1
SP0503BAHT
D1
USB_N
USB_P
FB2 BLM18SG121TN1D
F1 1A
+5V
C6 0.01uF
C8 0.1uF
FB1 BLM31PG121SN1L
F2 2A
5V_TABLET
C5 0.01uF
C7 0.1uF
F1 -> Hold=1A Trip=2A Rmax=0.23
F2 -> Hold=2A Trip=3.5A Rmax=0.08
FB1 -> 3A 25mΩ 120Ω@100MHz
FB2 -> 3.5A 25mΩ 120Ω@100MHz
USB
U3
LR1107
VIN
VOUT
C16 10uF
+3V3
TP4
3V3
C18 100uF
C20 100uF
Est. Consumptions:
Idle: 160 mA
Max: 1050 mA (worst case!)
3V3 LDO
+5V
L1 10uH
3.5-5.0V
5V_CLEAN
U1
XC6228D332VR-G
CE
C2 10uF
C3 10uF
C4 0.1uF
3V3_CLEAN
TP1
3V3_C
C9 0.1uF
C10 10uF
FB3
BLM15PD121SN1B
TP2
3V3_S
3V3_SENSOR
C11 0.1uF
C13 0.1uF
FB4
BLM15PD121SN1B
3V3_LDO1V8
U2
XC6228D182VR-G
TP3
1V8_S
1V8_SENSOR
C12 0.1uF
C14 10uF
C15 0.1uF
C17 0.1uF
C19 10uF
SENSOR FILTERED LDO
All resistors are 1% unless otherwise mentioned.
Ezlo
Sheet: /Power-Interface/
File: power.kicad_sch
Title: Ezlo Advanced Thermostat
Size: A4
Date: 2022-12-19
Rev: B
KiCad E.D.A. kicad (6.0.5)
Id: 2/15

RADAR SENSOR



Radar sensor sits externally on it's own PCB
and is connected via FPC to the maimboard.

4-in-1 VOC Humidity Pressure Temeperature



<- SDO is I2C address bit 0

CSB -> VDDIO = I2C active
CSB -> GND = SPI active
I2C ADDR = BIN 1110111 HEX 0x77
I2C pull-up resistors are on the ESP32 sheet.

BSEC: Bosch Sensortec Environmental Cluster
BSEC precisely performs several calculations outside the device such as ambient air temperature, ambient relative humidity, pressure and air quality (IAQ) level.
https://www.bosch-sensortec.com/software-tools/software/bsec/
https://github.com/BoschSensortec/BSEC-Arduino-library

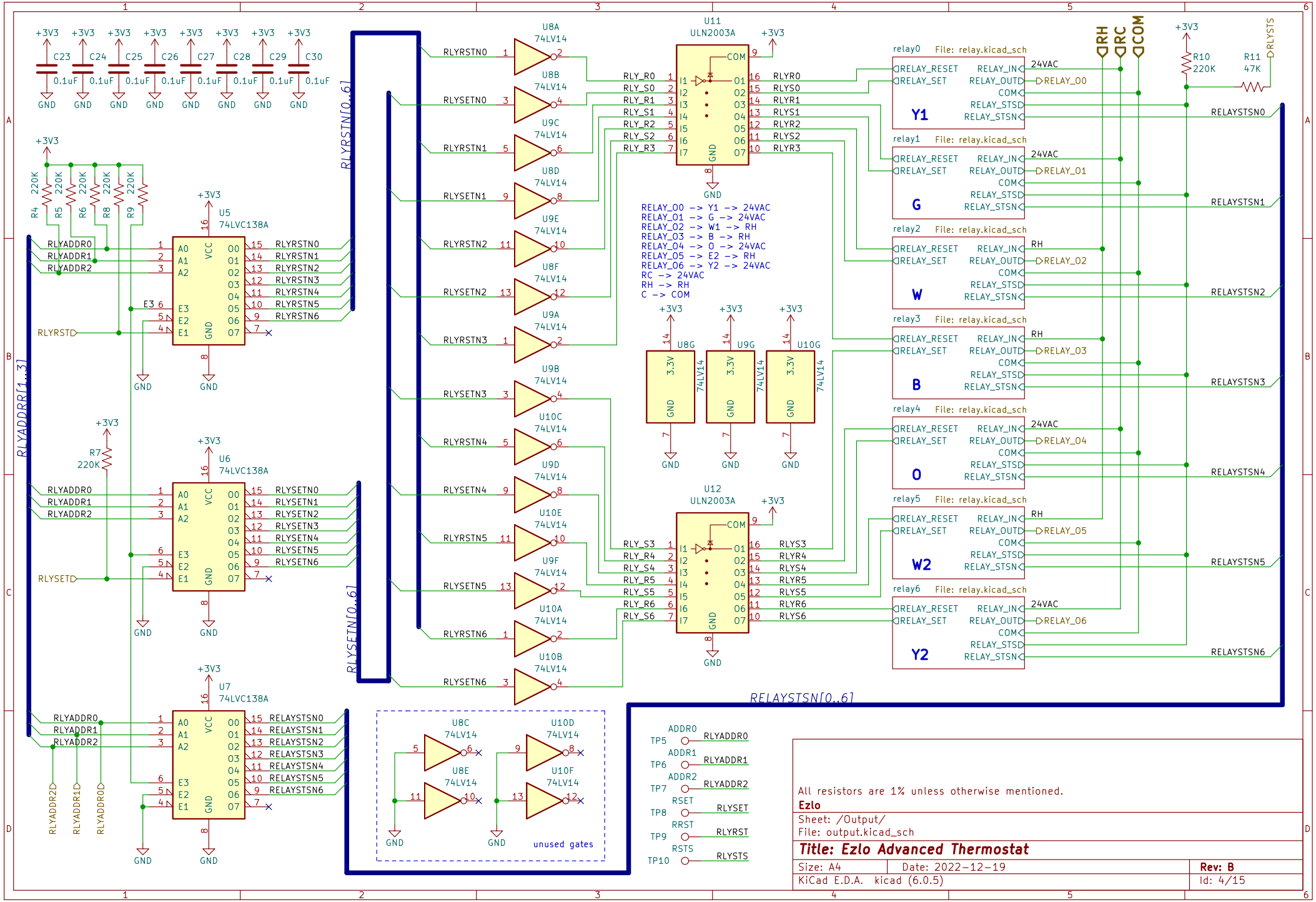
All resistors are 1% unless otherwise mentioned.

Ezlo

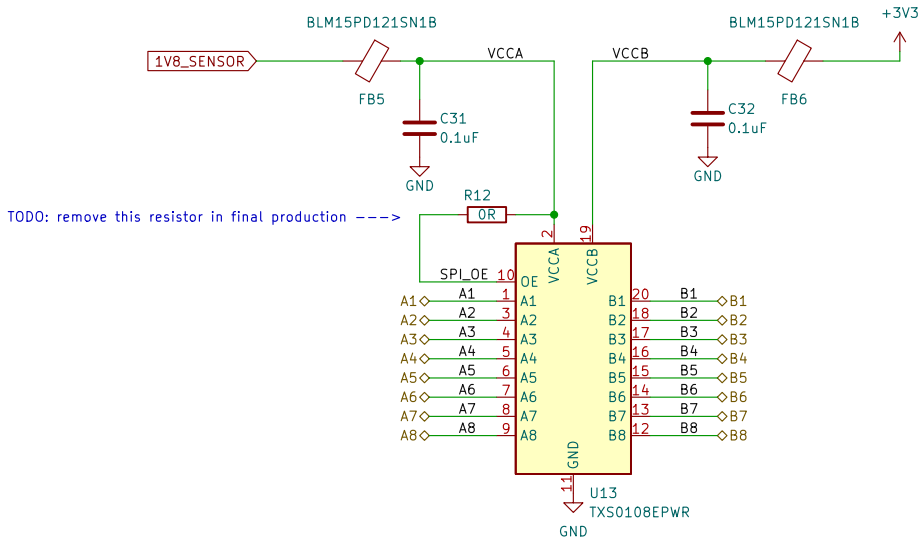
Sheet: /Sensors/
File: sensors.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 3/15


C23 C24 C25 C26 C27 C28 C29 C30
0.1uF
+3V3
GND
R4 R5 R6 R8 R9
220K
U5
U6
U7
74LVC138A
RLYADDR0
RLYADDR1
RLYADDR2
RLYRSTD
RLYSETD
RLYADDR2D
RLYADDR1D
RLYADDR0D
RLYSTN[0..6]
RLYSETN[0..6]
RLYADDR[1..3]
RLYRSTN0
RLYRSTN1
RLYRSTN2
RLYRSTN3
RLYRSTN4
RLYRSTN5
RLYRSTN6
RLYSETN0
RLYSETN1
RLYSETN2
RLYSETN3
RLYSETN4
RLYSETN5
RLYSETN6
RELAYSTSN0
RELAYSTSN1
RELAYSTSN2
RELAYSTSN3
RELAYSTSN4
RELAYSTSN5
RELAYSTSN6
U8A U8B U9C U8D U9E U8F U9A U9B U10C U9D U10E U9F U10A U10B
74LV14
U8G U9G U10G
3.3V
U11 U12
ULN2003A
RLY_R0 RLY_S0 RLY_R1 RLY_S1 RLY_R2 RLY_S2 RLY_R3
RLY_S3 RLY_R4 RLY_S4 RLY_R5 RLY_S5 RLY_R6 RLY_S6
RLYR0 RLYS0 RLYR1 RLYS1 RLYR2 RLYS2 RLYR3
RLYS3 RLYR4 RLYS4 RLYR5 RLYS5 RLYR6 RLYS6
RELAY_00 -> Y1 -> 24VAC
RELAY_01 -> G -> 24VAC
RELAY_02 -> W1 -> RH
RELAY_03 -> B -> RH
RELAY_04 -> O -> 24VAC
RELAY_05 -> E2 -> RH
RELAY_06 -> Y2 -> 24VAC
RC -> 24VAC
RH -> RH
C -> COM
relay0 relay1 relay2 relay3 relay4 relay5 relay6
File: relay.kicad_sch
RELAY_RESET
RELAY_SET
RELAY_IN
RELAY_OUT
COM
RELAY_STSD
RELAY_STSN
Y1 G W B O W2 Y2
24VAC
RH
RELAY_00 RELAY_01 RELAY_02 RELAY_03 RELAY_04 RELAY_05 RELAY_06
RELAYSTSN0 RELAYSTSN1 RELAYSTSN2 RELAYSTSN3 RELAYSTSN4 RELAYSTSN5 RELAYSTSN6
RH RC COM
R10 220K
R11 47K
RLYSTS
U8C U8E U10D U10F
unused gates
ADDR0 ADDR1 ADDR2 RSET RRST RSTS
TP5 TP6 TP7 TP8 TP9 TP10
RLYADDR0 RLYADDR1 RLYADDR2 RLYSET RLYRST RLYSTS
All resistors are 1% unless otherwise mentioned.
Ezlo
Sheet: /Output/
File: output.kicad_sch
Title: Ezlo Advanced Thermostat
Size: A4
Date: 2022-12-19
Rev: B
KiCad E.D.A. kicad (6.0.5)
Id: 4/15

Level Shifter



1.4 V to 3.6 V on A port and 1.65 V to 5.5 V on B port (VCCA ≤ VCCB)!

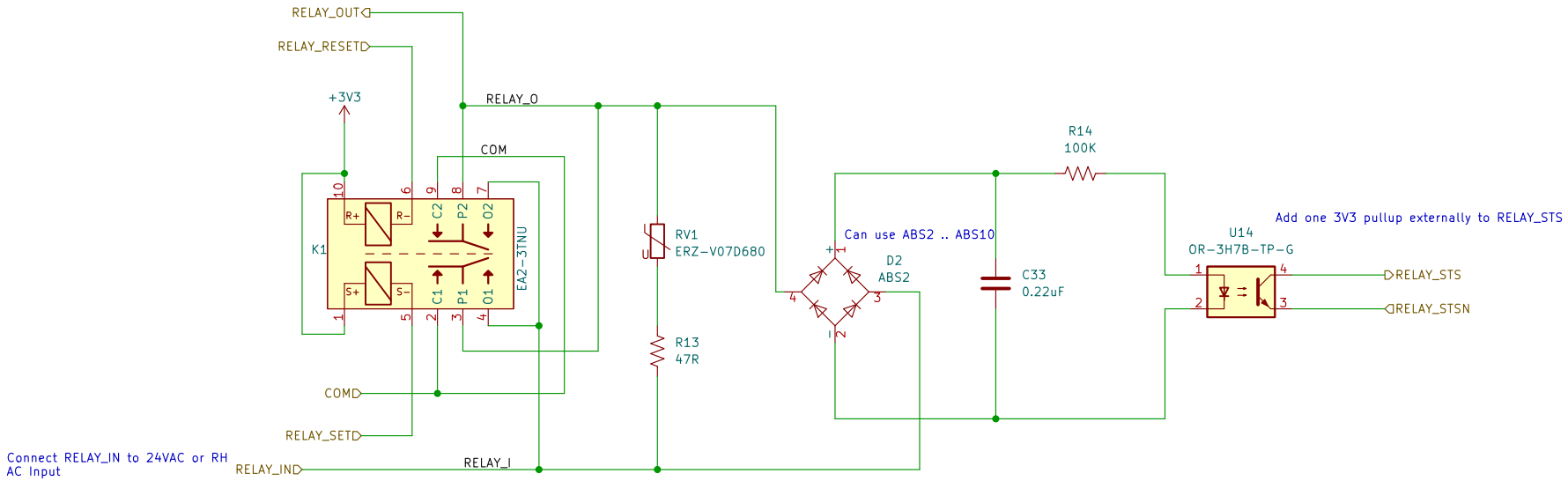
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Level shifter/
File: level_shift.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 5/15



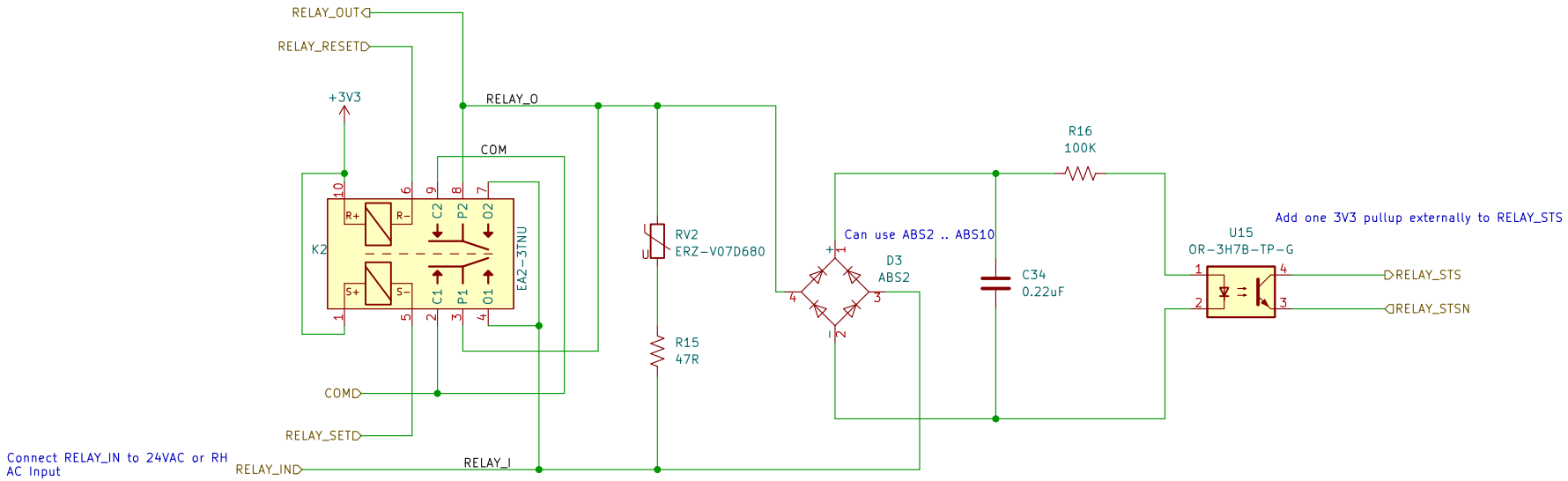
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay0/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 6/15



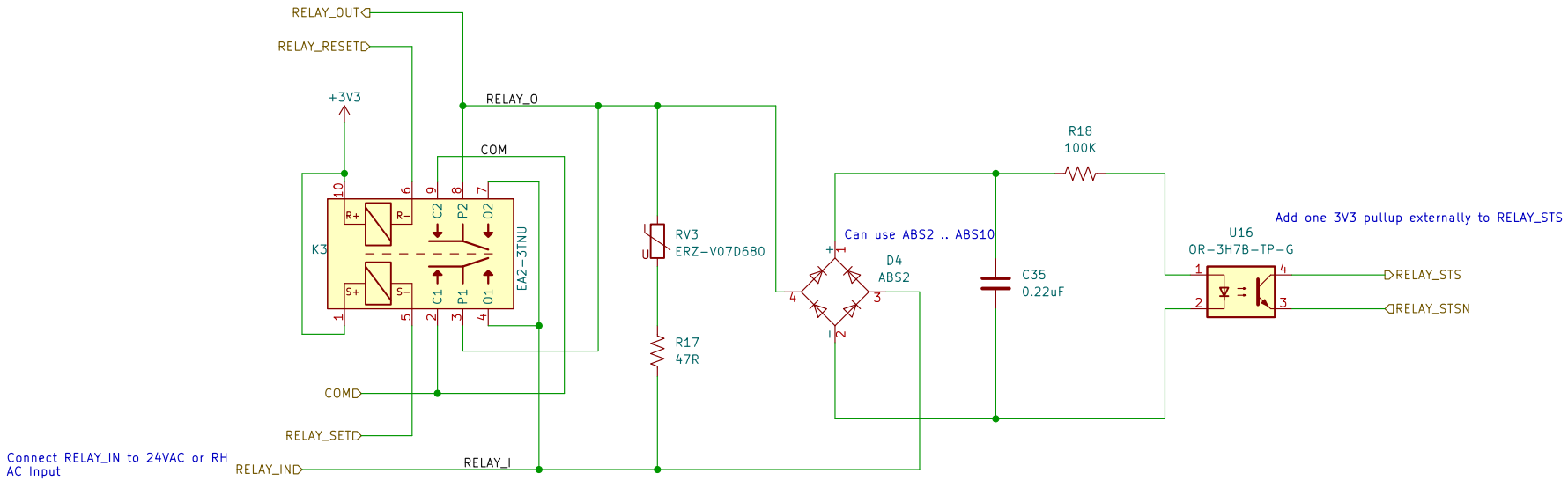
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay1/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 7/15



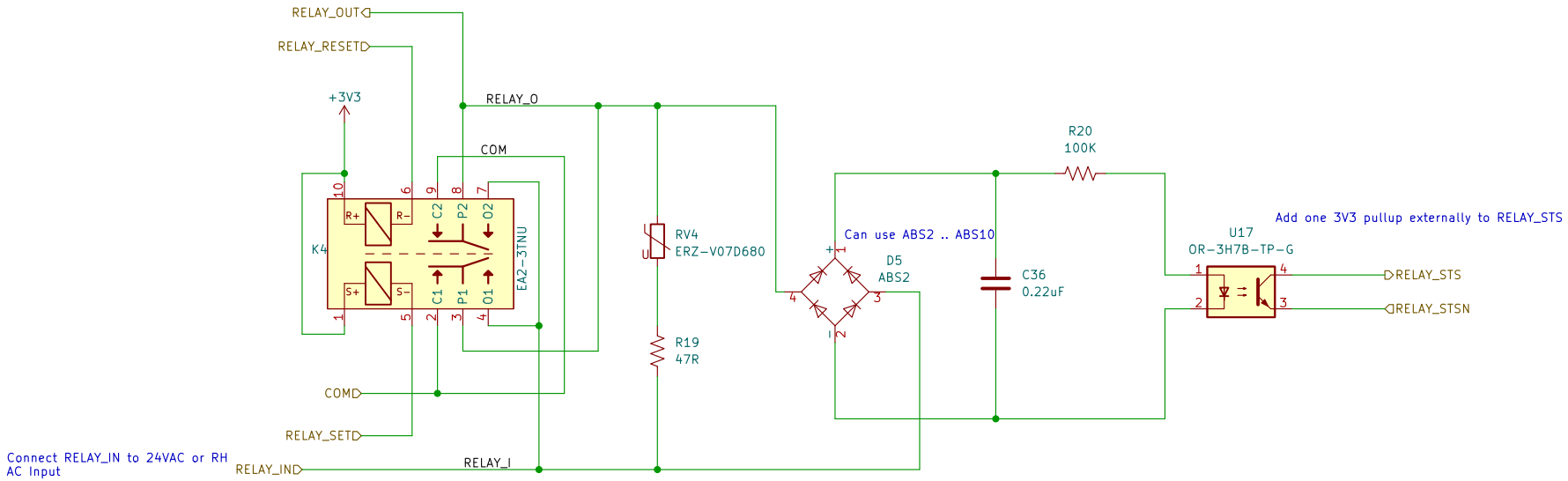
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay2/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 8/15



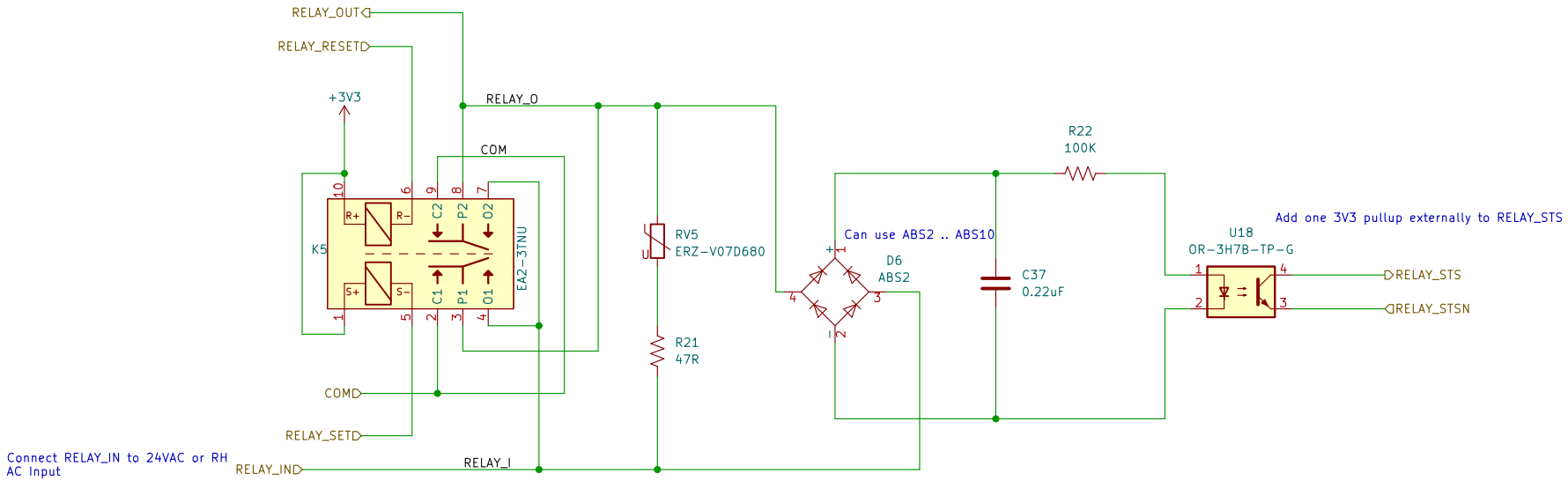
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay3/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 9/15



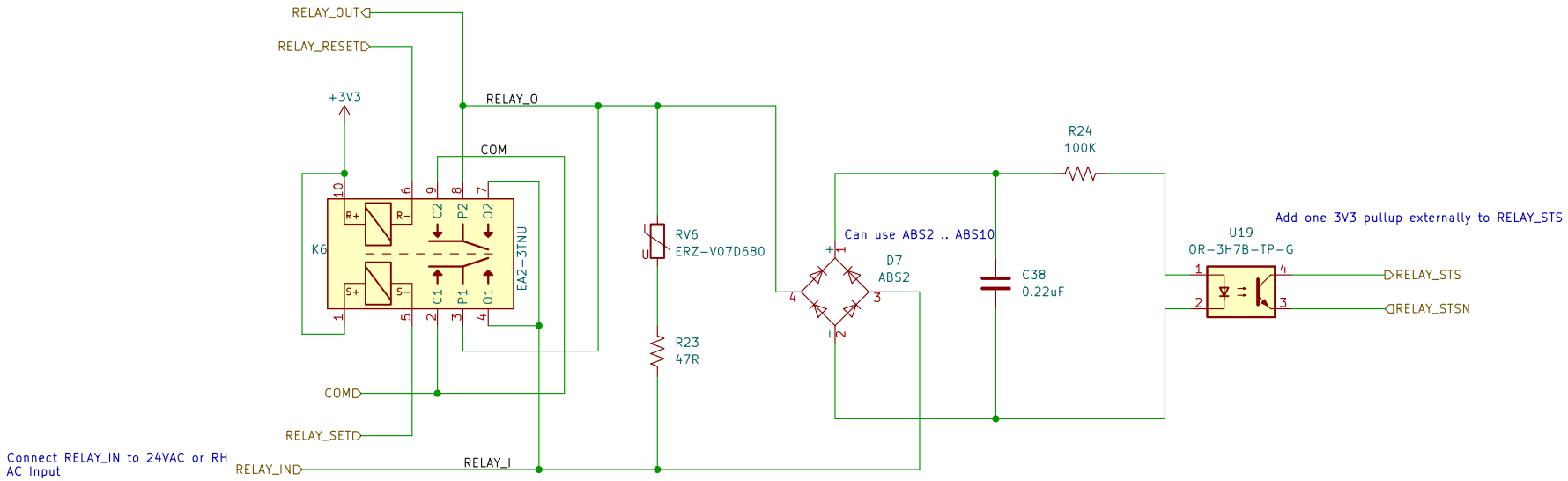
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay4/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 10/15



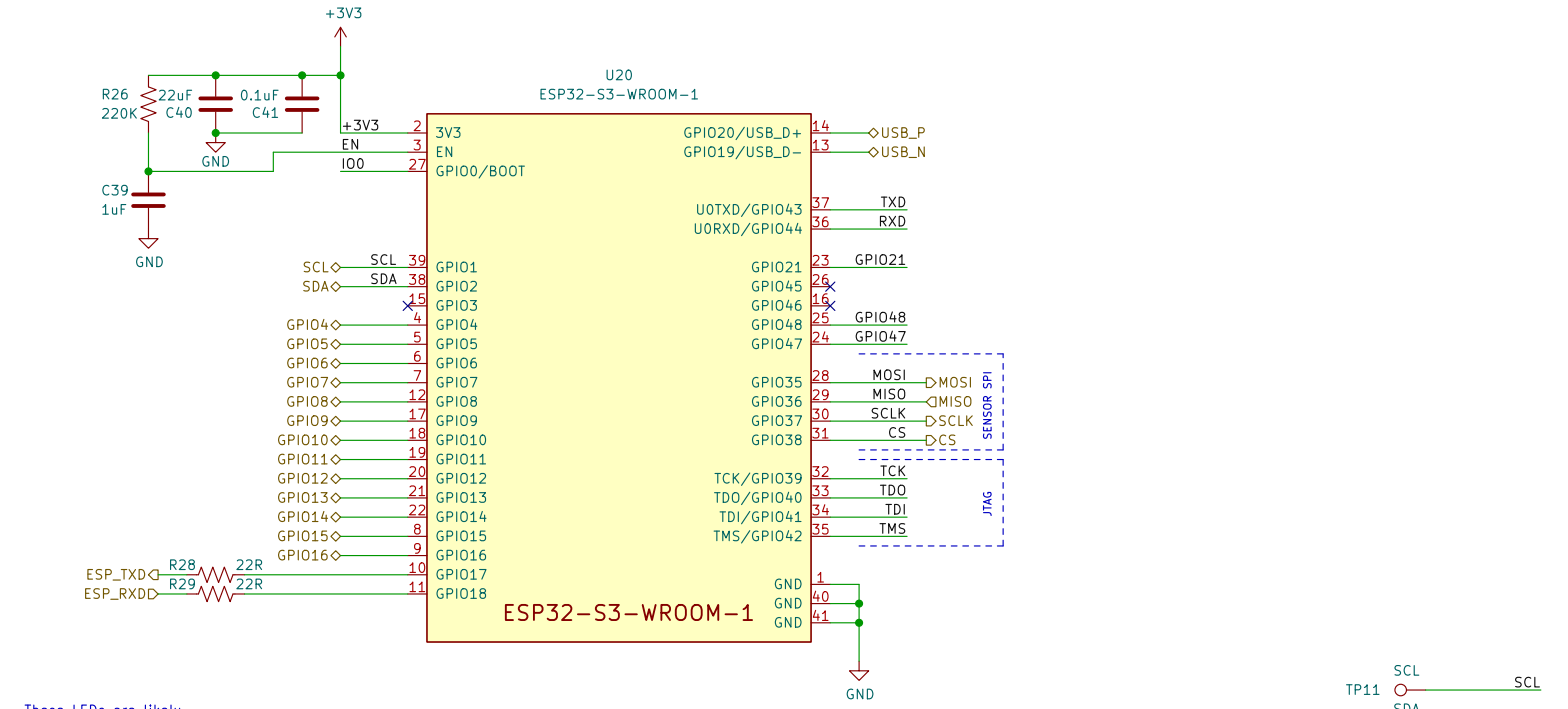
All resistors are 1% unless otherwise mentioned.

Ezlo

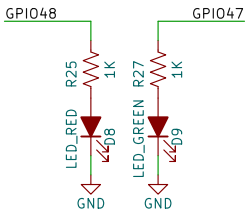
Sheet: /Output/relay5/
File: relay.kicad_sch

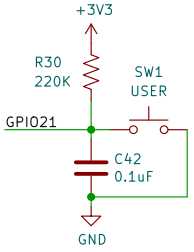
Title: Ezlo Advanced Thermostat

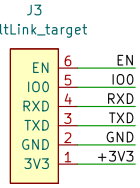
Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 11/15

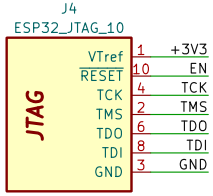


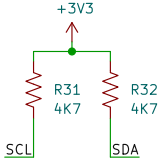
These LEDs are likely not needed in production!

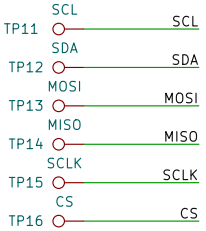












Strapping pins
- BOOT = IO0
- JTAG = IO3
- VSPI = IO45
- LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO33~37 are connected to SPIIO4 ~ SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO33~37 are also not recommended for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs, USB-JTAG will be disabled by the drivers.

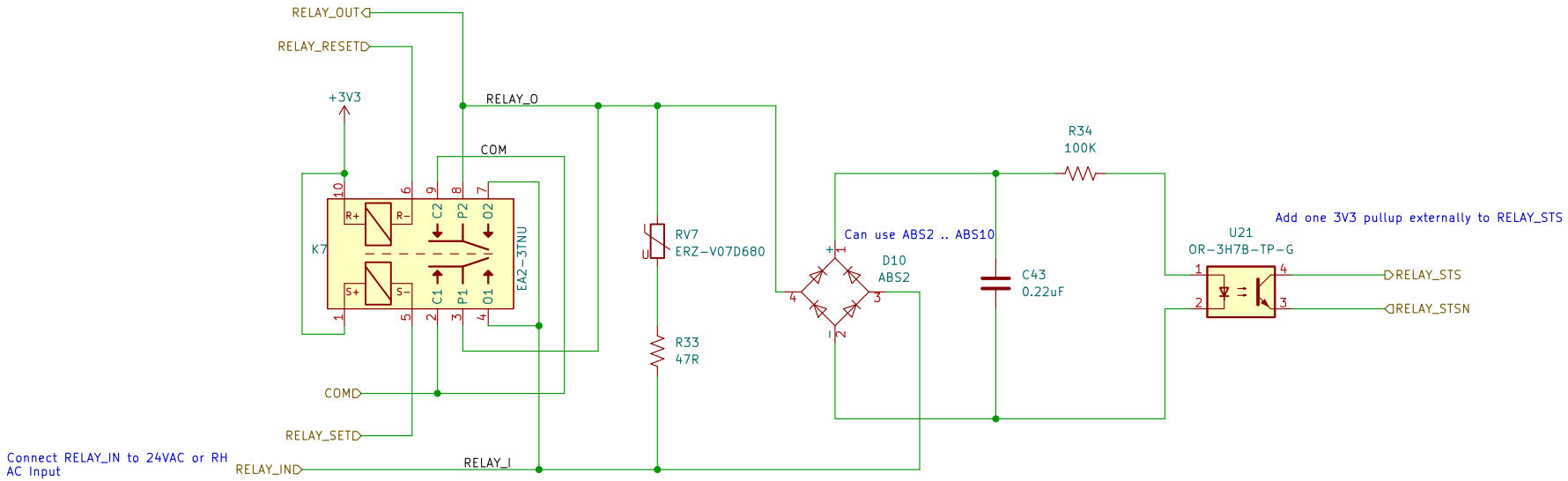
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /ESP32/
File: esp32.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 12/15



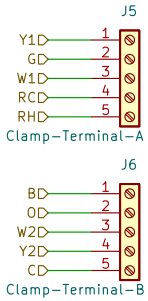
All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output/relay6/
File: relay.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 12/15



RELAY_O0 -> Y1
RELAY_O1 -> G
RELAY_O2 -> W1
RELAY_O3 -> B
RELAY_O4 -> O
RELAY_O5 -> W2
RELAY_O6 -> Y2
RC -> 24VAC
RH -> RH
C -> COM

All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Output-Connections/
File: output-connections.kicad_sch

Title: Ezlo Advanced Thermostat

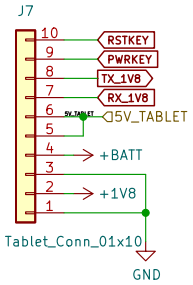
Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 13/15

TABLET INTERFACE

*interface with tablet 10 pins JST-SH 1A 50VDC rating, pinout contains:

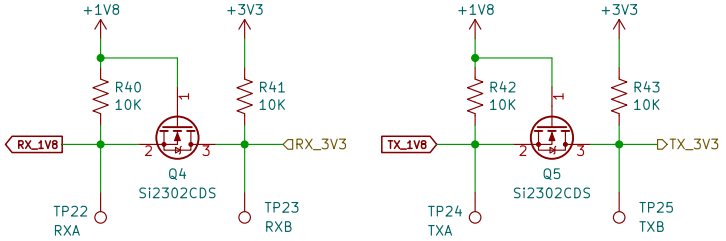
10. RESETKEY (output going to tablet, pulled high by tablet to 1V8, need to be pulled down via mosfet to GND)
9. PWRKEY (output going to tablet, pulled high by tablet to VBAT, needs to be pulled down via mosfet by ESP32)
8. TX (with voltage translation 1V8 > 3V3 so that ESP32 can talk to tablet processor)
7. RX (with voltage translation 1V8 > 3V3 so that ESP32 can talk to tablet processor)
6. VUSB (output going to tablet USB for power & charging)
5. VUSB (output going to tablet USB for power & charging)
4. VBAT (input coming from battery)
3. GND
2. 1V8 (input coming from tablet)
1. GND

Battery on tablet is not needed so VBAT is not used in this revision!

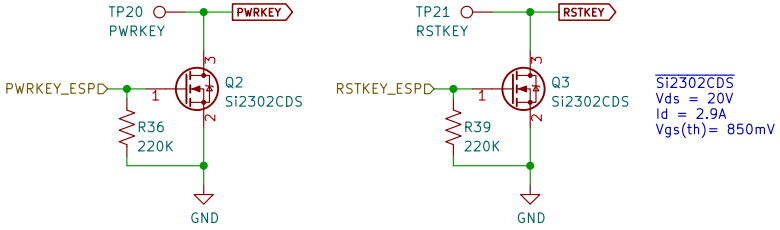


UART LEVEL CONVERSION

If a UART connection between Android tablet and ESP32 is not needed parts of this schematic will be DNP!



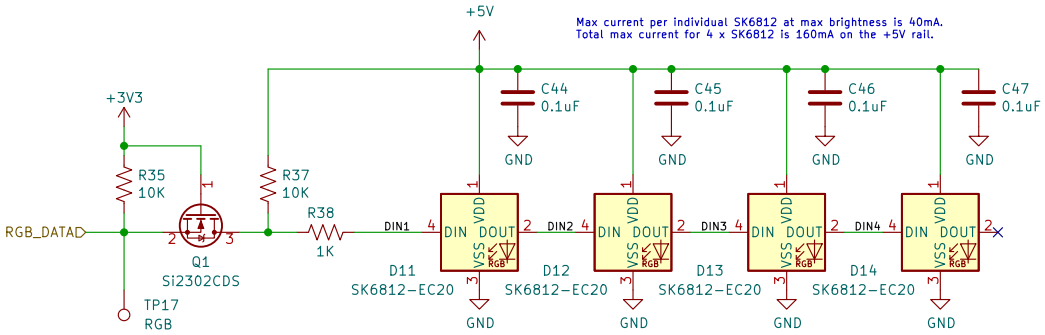
TABLET PWR & RST KEYS



Si2302CDS
Vds = 20V
Id = 2.9A
Vgs(th)= 850mV

PWRKEY (pulled high by tablet to VBAT)
RESETKEY (pulled high by tablet to 1V8)

RGB LEDs



TODO: Test behaviour when running on battery which may result in low VBAT!

TP18 5V_TABLET
5V_TABLET

TP19 +1V8
1V8

All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /misc/
File: misc.kicad_sch

Title: Ezlo Advanced Thermostat

Size: A4	Date: 2022-12-19	**Rev: B**
KiCad E.D.A. kicad (6.0.5)		Id: 14/15